

Demonstrating JumpMod: Haptic Backpack that Modifies Users' Perceived Jump

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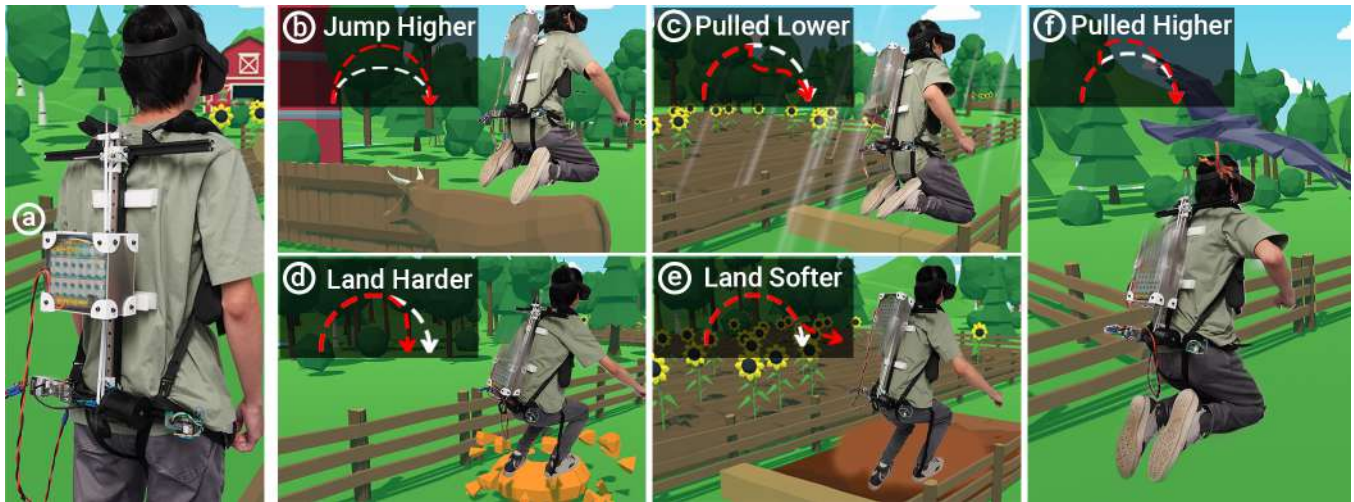


Figure 1: (a) We propose JumpMod, a wearable device that modifies the user's *perceived jump*. It achieves this by moving a weight along the user's back, causing an inertial force that modifies one's sense of vertical momentum. JumpMod renders five distinct sensations: a sense of (b) jumping higher; (c) being pulled lower; (d) landing harder or (e) softer; and (f) being pulled higher.

ABSTRACT

Abstract. Vertical force-feedback is extremely rare in mainstream interactive experiences. This happens because existing haptic devices capable of sufficiently strong forces that would modify a user's jump require grounding (e.g., motion platforms or pulleys) or cumbersome actuators (e.g., large propellers attached or held by the user). To enable interactive experiences to feature jump-based haptics without sacrificing wearability, we propose JumpMod, an untethered backpack that modifies one's sense of jumping. JumpMod achieves this by moving a weight up/down along the user's back, which modifies perceived jump momentum—creating accelerated & decelerated jump sensations. Our device can render five

distinct effects: jump higher, land harder/softer, pulled higher/lower, which we demonstrate in a jump-based VR experience.

KEYWORDS

jumping, full-body, haptics, virtual reality, wearable, backpack

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1 INTRODUCTION

A decade ago, users saw a fundamental change in the way they interacted with digital experiences in their homes. With the introduction of motion sensing (e.g., *Kinect*), the user's whole body could now serve as an input, enabling a more bodily way to interface with computers. Today, many interactive experiences are taking advantage of this, from interactive sports (e.g., laser-tag) to physical VR experiences (e.g., *BeatSaber* or VR fitness).

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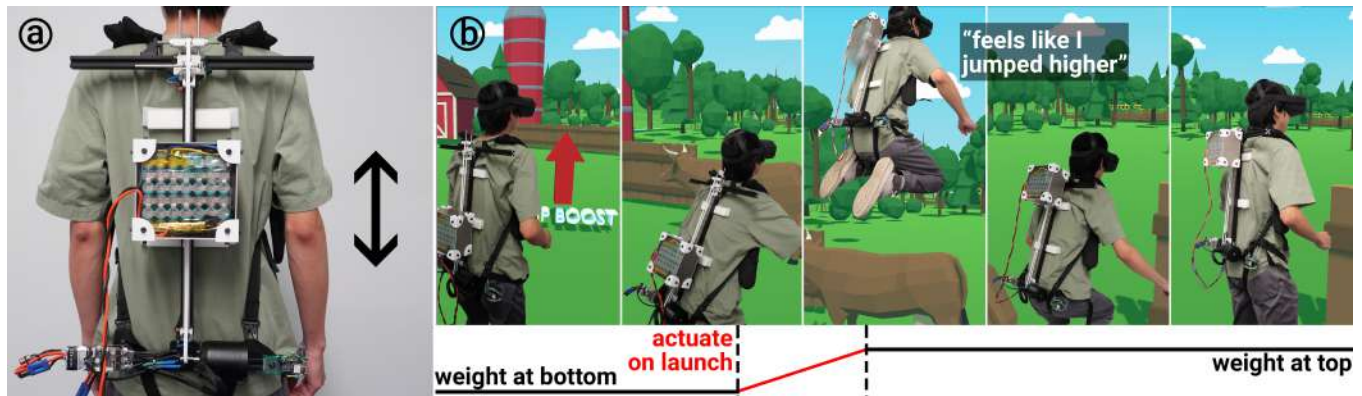


Figure 2: (a) *JumpMod* creates vertical haptic feedback by physically moving a weight up/down. It can create (b) the experience of jumping higher (during a power-up) by moving the weight up in sync with the *launch* phase of a jump, as depicted by the plot.

However, enabling interactive devices to move the user’s *entire body* (i.e., body-scale force-feedback) has proven challenging. This is primarily due to the large forces required to move the user’s body mass. Moreover, it is arguably easier to achieve force-feedback in the horizontal plane since a haptic device needs only to overcome the inertia and friction, which might explain the broader proliferation of interactive treadmills [7, 12] or walkable floors [8, 15] when compared to vertical force-feedback devices. A vertical force-feedback device needs to additionally overcome gravity (i.e., push against the user’s whole-body weight).

When researchers and practitioners (e.g., theme-parks or VR arcades designers) need to vertically actuate the user’s *entire* body, they usually resort to installing large actuators in the environment, such as motion platforms [1, 13], mechanically-actuated pulleys [5, 19], wind tunnels [3, 9] or attaching powerful propellers to the user’s body [14, 17]. While these are effective since their output force is large, they *limit entirely users’ mobility*—these devices are at odds with users’ home setups (e.g., free-walking VR experiences [6, 11]) and are at odds with experiences where users are highly mobile (e.g., interactive sports [2, 10]). Instead, to enable more mobile contexts, a vertical body-scale actuator that is *untethered and wearable* would prove ideal.

To tackle this and expand the portability of vertical force-feedback, we propose *JumpMod*, an untethered backpack that modifies the user’s perceived jump. Our device, depicted in Figure 1, achieves vertical effects by moving a weight along the user’s back, even in mid-air. The key behind *JumpMod* is that its actuator does not physically lift the user—as this would require motors and infrastructure 10x larger/heavier, preventing our small form-factor—instead, *JumpMod* creates only a *sufficiently* strong inertial force that users can feel, which *modifies* their *perceived* jump.

2 WALKTHROUGH

To illustrate the applicability of our device, we demonstrate it with an example of a jumping-based experience in VR. In this walkthrough, our VR user is wearing our untethered backpack and a wireless VR headset. This VR experience is set in a farm scene shown in Figure 2 (b), and the game is played by running and

jumping in place to avoid obstacles and collect points by smashing pumpkins.

Jumping higher. Our user is running in place and attempting to jump over obstacles; however, when they attempt to jump over a cow, they fail—they lose speed and must try again. On the second attempt, depicted in Figure 2 (a), they collect a jump boost power-up. As the user jumps again, our device actuates its weight *up* in sync with the user’s *launch* from the ground (our device takes only 20ms to detect any jump phase with high accuracy). This extra inertial force from the moving weight creates a sense of *jumping higher*. As a result, not only do they *see* their avatar jumping over the cow (as typical in jumping-based VR experiences, we also scale up the avatar’s jump), they *feel* this matches the *physical sensation* of jumping higher.

Pulling down. Figure 3 (a) illustrates our user jumping over the next obstacle, a haystack. This time when the user is still mid-jump, and about to land, the wind blows harder and pulls the user down (the user sees and hears the wind effects too). To generate this “pulling” sensation, our device moves its weight *down* as the user *lands*.

Landing softer. Figure 3 (b) also illustrates our user landing on mud rather than on solid ground as before. To render this “softer” landing from the mud, our device moves its weight *up* when the user *lands*.

Landing harder. Next, our user spots a pumpkin in Figure 4 (a), which they need to smash for points. They collect a force power-up that enables smashing pumpkins. To render this sensation of “landing harder” on the pumpkin, our device moves the weight *down* when the user *descends* (past the apex, the highest point of their jump).

Resetting the weight’s position. So far, our VR experience was designed with all effects in a sequence that left the device’s weight in the place where the next effect was required. But this is not necessary; there are several ways that a designer can use to reset the weight’s position, ranging from options that conserve realism to those that provide speed: (1) planning the order of the effects so no reset is needed, (2) slowly driving the weight so that user cannot perceive it, (3) associating the reset with other in-game events

(e.g., an earthquake that triggers back-and-forth vibrations, but also move the weight to the intended position), and (4) moving it rapidly before the next event. In this particular experience, the designer chose to look ahead to the next obstacle and, if needed, always slowly reset the weight to the required start position (inspired by *HapticTurk*'s lookahead for the next haptic events [4]). As the next event requires the weight to start at the top but is currently at the bottom, the VR application sends a "5s reset up" instruction to our device, which brings the weight up in 5s. To ensure the weight is at the correct position, our device implements a positional PID controller (see *Implementation* details).

Pulled higher: Finally, Figure 4 (c) depicts the user having collected all the pumpkins and is about to jump over a final obstacle—they can already see the finish line. As they jump, a bird flies next to the user and pulls the user up to help them to the finish line. Our device renders this "pulled higher" by moving the weight *down* at the user's *launch*.

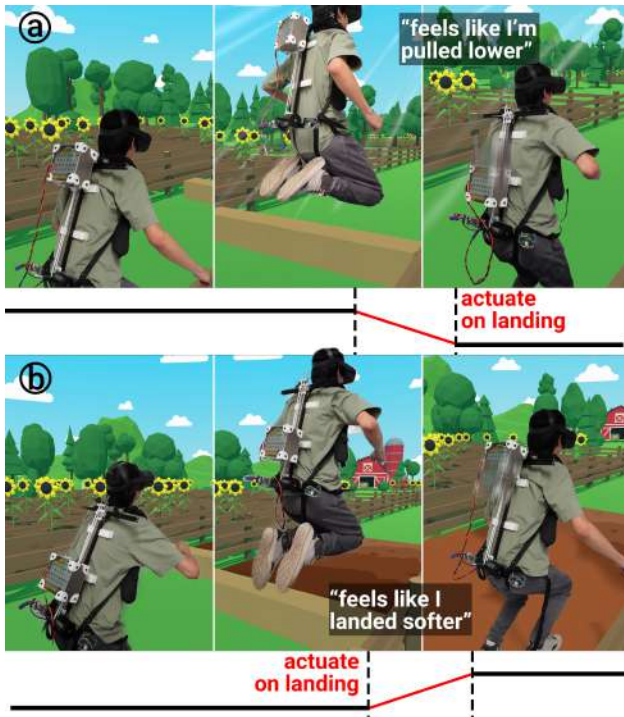


Figure 3: (a) *Pulled down* is rendered when the wind is blowing the user. (b) *Land softer* is rendered when the user lands in the mud.

3 IMPLEMENTATION

JumpMod is implemented as a self-contained haptic backpack (i.e., works on battery and is wireless)—to accelerate researchers interested in reproducing this device, all designs and code are available and open-sourced¹. Details on the implementation and characteristics of our device can be found in [16].

¹<https://lab.plopes.org/#JumpMod>

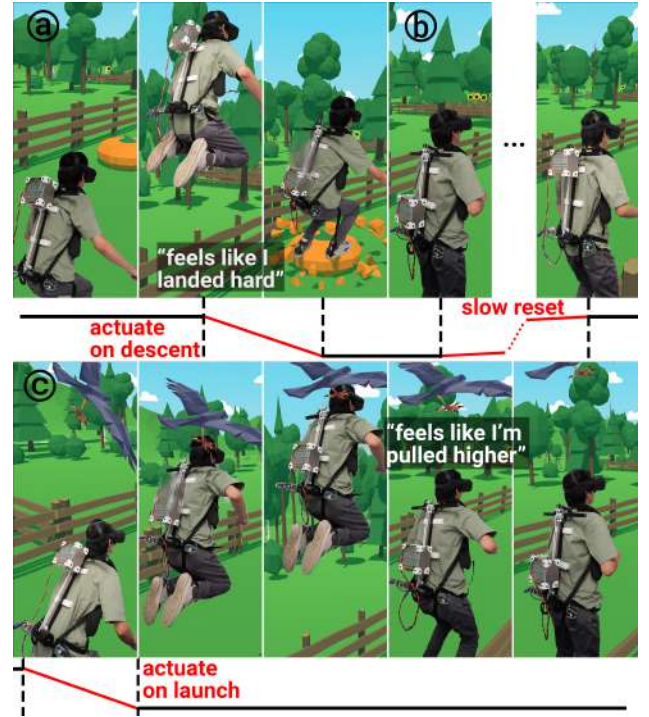


Figure 4: (a) *Landing harder* is rendered when the user lands on a pumpkin to smash it. (b) The weight slowly moves up to reset its position in preparation for the next effect. (c) *Pulled higher* is rendered when a bird assists the user during a jump.

3.1 Hardware

Our device's hardware can be broken down into four main categories: (1) structure, (2) battery as moving weight and source of power, (3) drive system, and (4) control.

Structure. The structure of the device is built around three 2020 aluminum extruded profiles, shaped to form the letter "T" as depicted in Figure 5(a). We made this structure into a backpack and improved its ergonomics by adding padded straps (Meister Backpack Straps) on the four corners as well as hard-foam padded supports along the aluminum profiles to spread the contact surface evenly on the user's back.

Battery as moving weight and as power. Typical weight displacement devices opt for weights made from dense materials (e.g., lead), which allows the weight to be as small as possible, and yet heavy. However, these devices still need to attach a power source—the larger the weight, the heavier batteries become. While this is not an issue for grounded devices, this is critical for comfort in wearables. Instead, in JumpMod, our battery has dual functionality: as weight *and* as a power source. To create a battery that satisfied both requirements perfectly, i.e., one that weighed approximately ~2kg and provided fast current discharge for our motor's operation, we created a custom battery by manually packing cylindrical Li-ion cells. Our resulting custom 24V 7Ah battery is made from 42 cylindrical Li-ion cells (Anhui Ruikema Li-ion ICR18650-20P5C, 3.7V, and 2000mAh) in a 6S7P configuration, weighing 2kg. It is

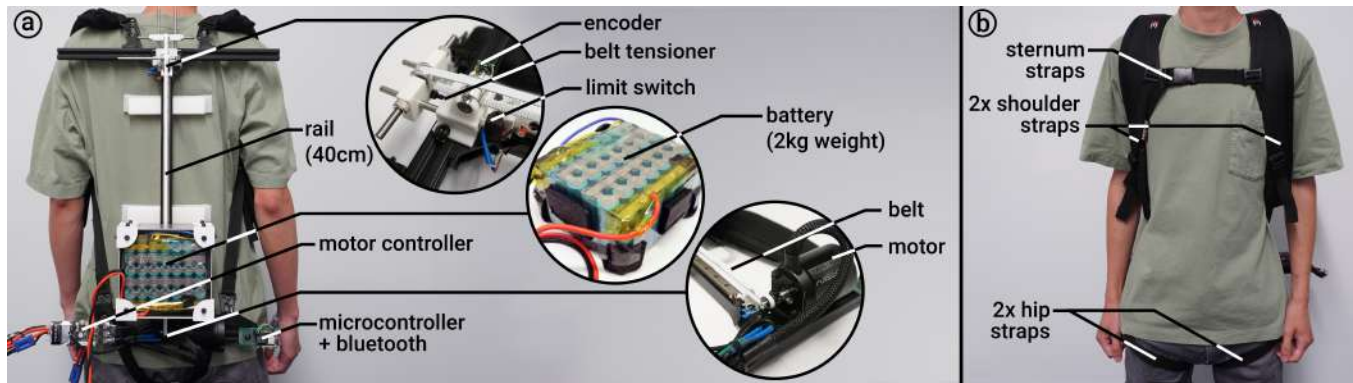


Figure 5: (a) JumpMod is untethered and wearable; it features: a custom battery that doubles as a 2kg weight, a driving belt system with an encoder, and a microcontroller with Bluetooth. (b) Four anchor points and a sternum strap secure the backpack to the user.

enclosed in a box we engineered from five 2mm waterjet-cut aluminum plates and a 3 mm acrylic plate, connected via 3D printed PLA corner brackets.

Drive system. JumpMod is actuated by a brushless DC motor (Flipsky Outrunner 6374, 140kv, 980g, and 3.5kW max power output) controlled by an electronic speed controller (Flipsky FSESC V4.12 50A). This motor was chosen for its torque output (max 8Nm) in a small and lightweight form factor (980g). Our motor drives the weight using a timing belt pulley (6mm steel core reinforced 2GT) along a linear rail guide (MGN12H 400mm). Since this weight is moved relatively fast (707ms to actuate the weight from end-to-end), the steel core belt is essential to maintain the tension as well as avoid degradation over time (compared to the more commonly found nylon core belts). A custom 3D-printed PLA tensioner resides at the other end of the belt, allowing us to ensure high tension to maximize the power transfer from the motor to moving the weight while avoiding slipping.

Encoder. Our haptic effects are based on the inertia generated by moving the weight and not by any vibrations generated by the weight hitting the ends of the rail. As such, to move and stop the weight accurately, we implemented an encoder that determines the belt's movement. We employ a rotary magnetic encoder (Pololu Magnetic Encoder Pair Kit for Micro Metal Gearmotors) attached to an idle pulley, which keeps track of the position of the weight.

Controlling the weight's position. This encoder closes the loop and controls the weight position precisely by ensuring the weight reaches target positions. To control this accurately, we implemented a PID controller that maximizes the speed and prevents parasitic movements (overshoots or oscillations).

Microcontroller. The logic in our device is implemented by a Seeeduino XIAO nRF52840 BLE Sense. This was chosen because of its embedded sensors and, more importantly, its built-in Bluetooth. The microcontroller is powered by the electronic speed controller's 5V and communicates to it via UART using the VESC library [18].

Safety. Finally, two limit switches are located at the ends of the rail. If these are triggered, the microcontroller stops the motor. Moreover, since microcontrollers can also be affected by faults, we

added metallic physical stoppers at the ends of the rail, which in the event of a limit switch failure, the weight stays inside the rail.

3.2 Jumping detection algorithm

We detect when users jump using a simple algorithm based on absolute head position via an external tracking system using the HTC VIVE Lighthouse V2 tracking system and its headset. We implemented it in Unity3D communicates wirelessly to our backpack via Bluetooth (68.5ms of latency). At the start of tracking, our jump detection algorithm records the user's height as a reference. In runtime, whenever the user's position is below their height by 7.5cm, and there is no jump in progress, the algorithm determines that the user started crouching in preparation for the jump, and their position is currently decreasing. Now, our algorithm is ready to track four jump phases: *launch*, *ascent*, *descent*, and *landing*. First, our algorithm triggers the *launch* phase when the user's head position starts increasing again (extending after crouching). Second, our algorithm triggers the *ascent* phase when the user's height surpasses their reference height (their feet start to lift from the floor). Third, our algorithm triggers the *descent* phase when the user's height stops increasing and starts decreasing (they pass the jump's apex). Finally, our algorithm triggers the *landing* phase when the user's height crosses their reference height (the feet are touching the floor). Our algorithm is able to differentiate a real jump (user's feet left the ground) from a fake jump (user intended to jump but decided to abort keeping their feet on the ground) with 94% accuracy.

4 CONCLUSIONS AND FUTURE WORK

We demonstrated a wearable device, which we call *JumpMod*, that modulates the user's perceived jump. JumpMod achieves this by moving a weight along the user's back, which causes an inertial force that modulates the user's perception of their own vertical momentum. Unlike existing approaches (e.g., wearing or holding large propellers to gain vertical momentum or sitting or standing on grounded motion platforms), JumpMod modifies the user's perceived jump using its small form factor, small enough that it is worn as a backpack—it is completely untethered, users can walk around

freely and do not need to hold any handles to experience its haptic effects.

JumpMod can render five distinct sensations: (1) a sense of jumping higher, (2) a sense of being pulled lower, (3) a sense of landing harder, (4) a sense of landing softer, and (5) a sense of pulled higher. We demonstrated these effects in a jump-based VR experience.

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